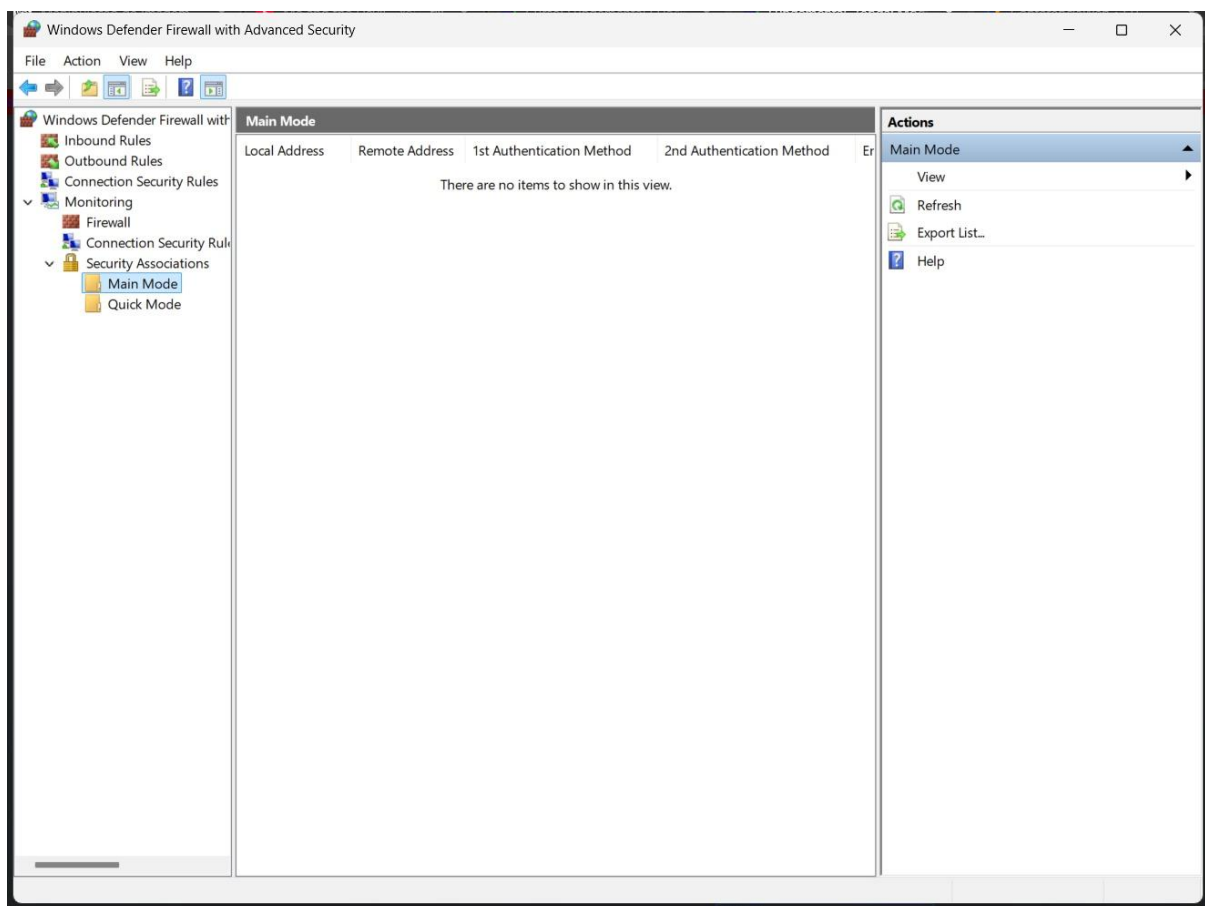


Atividade 22

Print da atividade 11.6: Auditoria de associações de segurança ativas no Firewall do Windows em modo Main Mode



Print da atividade 11.7: Criação de regra no iptables para bloqueio de saída de tráfego para

um endereço IP externo específico

```
root@Cleydson: ~
Sessão Ações Editar Exibir Ajuda
(cleydson@Cleydson)-[~]
$ xrandr --output HDMI-1-0 --mode 1920x1080 --rotate right --right-of eDP-1
(cleydson@Cleydson)-[~]
$ sudo -i
[sudo] senha para cleydson:
(root@Cleydson)-[~]
# iptables -A INPUT -s globo.com.br -j DROP
(root@Cleydson)-[~]
# iptables -L
Chain INPUT (policy ACCEPT)
target     prot opt source                destination
DROP      all  --  186-192-83-5.prt.globo.com anywhere

Chain FORWARD (policy DROP)
target     prot opt source                destination
DOCKER-USER all  --  anywhere              anywhere
DOCKER-ISOLATION-STAGE-1 all  --  anywhere              anywhere
ACCEPT     all  --  anywhere              anywhere             ctstate RELATED,ESTABLISHED
DOCKER     all  --  anywhere              anywhere
ACCEPT     all  --  anywhere              anywhere
ACCEPT     all  --  anywhere              anywhere

Chain OUTPUT (policy ACCEPT)
target     prot opt source                destination

Chain DOCKER (1 references)
target     prot opt source                destination

Chain DOCKER-ISOLATION-STAGE-1 (1 references)
target     prot opt source                destination
DOCKER-ISOLATION-STAGE-2 all  --  anywhere              anywhere
RETURN     all  --  anywhere              anywhere

Chain DOCKER-ISOLATION-STAGE-2 (1 references)
target     prot opt source                destination
DROP      all  --  anywhere              anywhere
RETURN     all  --  anywhere              anywhere

Chain DOCKER-USER (1 references)
target     prot opt source                destination
RETURN     all  --  anywhere              anywhere
(root@Cleydson)-[~]
# iptables -D INPUT -s globo.com.br -j DROP
(root@Cleydson)-[~]
# iptables -L
Chain INPUT (policy ACCEPT)
target     prot opt source                destination

Chain FORWARD (policy DROP)
target     prot opt source                destination
DOCKER-USER all  --  anywhere              anywhere
DOCKER-ISOLATION-STAGE-1 all  --  anywhere              anywhere
ACCEPT     all  --  anywhere              anywhere             ctstate RELATED,ESTABLISHED
DOCKER     all  --  anywhere              anywhere
ACCEPT     all  --  anywhere              anywhere
ACCEPT     all  --  anywhere              anywhere

Chain OUTPUT (policy ACCEPT)
target     prot opt source                destination

Chain DOCKER (1 references)
target     prot opt source                destination

Chain DOCKER-ISOLATION-STAGE-1 (1 references)
target     prot opt source                destination
DOCKER-ISOLATION-STAGE-2 all  --  anywhere              anywhere
RETURN     all  --  anywhere              anywhere

Chain DOCKER-ISOLATION-STAGE-2 (1 references)
target     prot opt source                destination
DROP      all  --  anywhere              anywhere
RETURN     all  --  anywhere              anywhere

Chain DOCKER-USER (1 references)
target     prot opt source                destination
RETURN     all  --  anywhere              anywhere
(root@Cleydson)-[~]
#
```

Print da atividade 11.8: Implementação de redirecionamento de tráfego (NAT) para o

sequestro de requisições DNS

```
(root@Cleydson)-[~]
# iptables -t nat -A OUTPUT -p udp --dport 53 -j DNAT --to-destination 127.0.0.1:53

(root@Cleydson)-[~]
# iptables -t nat -L
Chain PREROUTING (policy ACCEPT)
target    prot opt source                destination
DOCKER    all  --  anywhere               anywhere           ADDRTYPE match dst-type LOCAL

Chain INPUT (policy ACCEPT)
target    prot opt source                destination

Chain OUTPUT (policy ACCEPT)
target    prot opt source                destination
DOCKER    all  --  anywhere               !127.0.0.0/8
DNAT      udp  --  anywhere               anywhere           ADDRTYPE match dst-type LOCAL
                                         udp dpt:domain to:127.0.0.1:53

Chain POSTROUTING (policy ACCEPT)
target    prot opt source                destination
MASQUERADE all  --  172.17.0.0/16          anywhere

Chain DOCKER (2 references)
target    prot opt source                destination
RETURN    all  --  anywhere               anywhere

(root@Cleydson)-[~]
# iptables -t nat -D OUTPUT -p udp --dport 53 -j DNAT --to-destination 127.0.0.1:53

(root@Cleydson)-[~]
# iptables -t nat -L
Chain PREROUTING (policy ACCEPT)
target    prot opt source                destination
DOCKER    all  --  anywhere               anywhere           ADDRTYPE match dst-type LOCAL

Chain INPUT (policy ACCEPT)
target    prot opt source                destination

Chain OUTPUT (policy ACCEPT)
target    prot opt source                destination
DOCKER    all  --  anywhere               !127.0.0.0/8           ADDRTYPE match dst-type LOCAL

Chain POSTROUTING (policy ACCEPT)
target    prot opt source                destination
MASQUERADE all  --  172.17.0.0/16          anywhere

Chain DOCKER (2 references)
target    prot opt source                destination
RETURN    all  --  anywhere               anywhere

(root@Cleydson)-[~]
#
```

Print da atividade 11.9: Estabelecimento de túnel VPN criptografado e validação da conexão segura entre hosts remotos

```
root@Cleydson: /usr/share/doc/openvpn/examples/sample-config-files

Sessão  Ações  Editar  Exibir  Ajuda

(root@Cleydson)-[~]
# cd /home/cleydson/Downloads

(root@Cleydson)-[/home/cleydson/Downloads]
# openvpn --config nome_do_arquivo.ovpn
Options error: In [CMD-LINE]:1: Error opening configuration file: nome_do_arquivo.ovpn
Use --help for more information.

(root@Cleydson)-[/home/cleydson/Downloads]
# cd /home/cleydson/Downloads

(root@Cleydson)-[/home/cleydson/Downloads]
# ls *.ovpn
ls: não foi possível acessar '*.ovpn': Arquivo ou diretório inexistente

(root@Cleydson)-[/home/cleydson/Downloads]
# find / -name "*.ovpn" 2>/dev/null

(root@Cleydson)-[/home/cleydson/Downloads]
# cd /usr/share/doc/openvpn/examples/sample-config-files/

(root@Cleydson)-[/usr/.../doc/openvpn/examples/sample-config-files]
# sudo openvpn --config client.conf
2026-03-25 20:03:21 Note: --cipher is not set. OpenVPN versions before 2.5 defaulted to BF-CBC as fallback when cipher negotiation failed in this case. If you need this fallback please add '--data-ciphers-fallback BF-CBC' to your configuration and/or add BF-CBC to --data-ciphers. E.g. --data-ciphers DEFAULT:BF-CBC
Options error: --ca fails with 'ca.crt': No such file or directory (errno=2)
Options error: --cert fails with 'client.crt': No such file or directory (errno=2)
2026-03-25 20:03:21 WARNING: cannot stat file 'client.key': No such file or directory (errno=2)
Options error: --key fails with 'client.key': No such file or directory (errno=2)
Options error: Please correct these errors.
Use --help for more information.

(root@Cleydson)-[/usr/.../doc/openvpn/examples/sample-config-files]
# openvpn --version
OpenVPN 2.7_rc4 x86_64-pc-linux-gnu [SSL (OpenSSL)] [LZO] [LZ4] [EPOLL] [PKCS11] [MH/PKTINFO] [AEAD] [DCO]
library versions: OpenSSL 3.5.4 30 Sep 2025, LZO 2.10
DCO version: 6.18.3+kali+2-amd64 #1 SMP PREEMPT_DYNAMIC Kali 6.18.3-1kali3 (2026-01-16)
Originally developed by James Yonan
Copyright (C) 2002-2025 OpenVPN Inc <sales@openvpn.net>
Compile time defines: enable_async_push=no enable_comp_stub=no enable_crypto_ofb_cfb=yes enable_dco=yes enable_dco_arg=yes
enable_debug=yes enable_dependency_tracking=no enable_dlopen=unknown enable_dlopen_self=unknown enable_dlopen_self_static=unknown
enable_dns_updown_by_default=yes enable_fast_install=needless enable_fragment=yes enable_iproute2=no enable_libtool_lock=yes
enable_lz4=yes enable_lzo=yes enable_maintainer_mode=no enable_management=yes enable_ntlm=yes enable_option_checking=no
enable_pam_dlopen=no enable_pedantic=no enable_pkcs11=yes enable_plugin_auth_pam=yes enable_plugin_down_root=yes enable_plugins=yes
enable_port_share=yes enable_selinux=no enable_shared=yes enable_shared_with_static_runtimes=no enable_silent_rules=no
enable_small=no enable_static=yes enable_strict=no enable_strict_options=no enable_systemd=yes enable_unit_tests=no
enable_werror=no enable_win32_dll=yes enable_wolfssl_options_h=yes enable_x509_alt_username=yes with_aix_soname=aix
with_crypto_library=openssl with_gnu_ld=yes with_mem_check=no with_openssl_engine=auto with_sysroot=no

(root@Cleydson)-[/usr/.../doc/openvpn/examples/sample-config-files]
#
```

Print da atividade 11.10: Configuração de regra de entrada no Firewall Linux para o bloqueio total de pacotes ICMP (Ping)


```
root@Cleydson: /usr/share/doc/openvpn/examples/sample-config-files

Sessão  Ações  Editar  Exibir  Ajuda

(root@Cleydson)-[/usr/.../doc/openvpn/examples/sample-config-files]
# hostname -I
192.168.15.77 172.17.0.1 2804:7f0:6540:12b0:6a48:80eb:2d39:49de

(root@Cleydson)-[/usr/.../doc/openvpn/examples/sample-config-files]
#

(root@Cleydson)-[/usr/.../doc/openvpn/examples/sample-config-files]
# ping -c 4 192.168.15.1
PING 192.168.15.1 (192.168.15.1) 56(84) bytes of data.
64 bytes from 192.168.15.1: icmp_seq=1 ttl=64 time=19.1 ms
64 bytes from 192.168.15.1: icmp_seq=2 ttl=64 time=36.7 ms
64 bytes from 192.168.15.1: icmp_seq=3 ttl=64 time=2.81 ms
64 bytes from 192.168.15.1: icmp_seq=4 ttl=64 time=21.3 ms

— 192.168.15.1 ping statistics —
4 packets transmitted, 4 received, 0% packet loss, time 3004ms
rtt min/avg/max/mdev = 2.805/19.988/36.716/12.018 ms

(root@Cleydson)-[/usr/.../doc/openvpn/examples/sample-config-files]
# iptables -A INPUT -p icmp --icmp-type echo-request -j DROP

(root@Cleydson)-[/usr/.../doc/openvpn/examples/sample-config-files]
# iptables -D INPUT -p icmp --icmp-type echo-request -j DROP

(root@Cleydson)-[/usr/.../doc/openvpn/examples/sample-config-files]
#
```